

# Solano County Health & Social Services Department



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## POLICY MEMORANDUM 6700

Effective Date: March 31, 2010  
Revised Date: April 24, 2017  
Review Date: April 24, 2019

### REVIEWED/APPROVED BY:

  
BRYN MUMMA, MD, MAS, EMS AGENCY MEDICAL DIRECTOR

  
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**SUBJECT: DESTINATION PROTOCOLS FOR AMBULANCES**

AUTHORITY: CALIFORNIA HEALTH & SAFETY CODE, DIVISION 2.5, 1797.222

### I. PURPOSE/POLICY:

To ensure that patients are transported to the appropriate facility and to establish the on-line medical control responsibilities of the designated base hospitals. This policy does not apply to trauma or ST Elevation Myocardial Infarction (STEMI) patients.

### II. DEFINITIONS

- A. **Emergency Patient:** A person who experiences an injury and who is subsequently transported by ambulance on an unscheduled basis.
- B. **Critical Patient:** An emergency patient who is at immediate risk to life or limb as evidenced by one or more of the following circumstances:
  - 1. Cardiogenic, hypovolemic, septic or anaphylactic shock.
  - 2. Severe respiratory distress and/or need for assisted ventilation.
  - 3. Coma or severe alteration of mental status.

4. Airway problems when measures available to the prehospital personnel are insufficient to establish and/or maintain a patent airway.
  5. Trauma patients as defined by Policy 6105, Solano County Prehospital Trauma Triage Plan.
  6. Active labor when delivery is imminent and complications are expected.
  7. Any patient who has a 12-lead EKG which is interpreted to show "acute MI" or "STEMI" per Policy 6609, ST Segment Elevation Myocardial Infarction.
  8. Any patient with the new onset of stroke symptoms that are less than three hours old.
  9. Any other patient that the Paramedic deems critical.
- C. **Non-Critical Patient:** An emergency patient who does not meet the criteria for "critical patient".
- D. **Patient Choice:** The chosen destination facility as stated by the patient or surrogate decision maker. If a patient is unable to communicate a choice the prehospital personnel shall use whatever other sources of information that might be available to indicate a pre-existing relationship. In the absence of a stated choice or indications of a usual provider, the patient is considered an undesignated patient.
- E. **Undesignated Patient:** A patient, or surrogate decision maker, who does not or cannot choose a destination facility.
- F. **Base Hospital:** A receiving hospital in Solano County which has contracted with the Solano County EMS Agency to provide on-line medical direction to ALS ambulances. The base hospitals in Solano County are:
1. North Bay Medical Center (NBMC)
  2. Kaiser Permanente Medical Center, Vallejo (KVH)
  3. Sutter Solano Medical Center (SSMC)
  4. VacaValley Hospital (VVH)
  5. Kaiser Foundation Medical Center, Vacaville (KVV)
- G. **Ambulance Zone A:** The cities of Vallejo and Benicia plus both directions of Interstate 80 (I-80) west of the American Canyon Road overcrossing, including the overcrossing. Zone A also includes the westbound lanes of I-80 between Red Top Road and the American Canyon Road overcrossing and both directions of Interstate 680 from Lake Herman Road up to but not including the Parish Road overcrossing.

### **III. GENERAL**

- A. No ambulance shall transport an emergency patient to any facility other than one licensed pursuant to Title 22, California Code of Regulations as having Basic or Comprehensive Emergency Medical Service.
- B. Patients are entitled to choose the hospital where their care is to be given. With all Solano County hospitals participating in giving on-line medical control, in most cases the base hospital guiding their prehospital care will be their chosen facility. With the exception of situations when the patient requests David Grant Medical Center (DGMC), the base hospital will also be the receiving hospital.
- C. All patient advisories and requests for physician orders or destination decisions shall be made to the base hospital using the established pre-recorded phone lines. In event of communication failure or inability to contact a physician at the receiving facility, prehospital personnel may contact another base hospital to obtain orders.
- D. In "Ambulance Zone A", SSMC and KVH are considered equidistant and will alternate in receiving undesignated patients by date. KVH will receive undesignated patients on the odd days of the month. SSMC will receive undesignated patients on even days of the month. The day of the call is determined by the time the call is dispatched. In the event of multiple undesignated patients during a single call, the patients will be divided equally among receiving facilities.
- E. Solano County EMS policy does not allow for ambulance diversion except in cases when a facility is incapacitated or temporarily unable to provide a vital service such as, but not limited to, CT scanning. These will be rare events and such redirection should be documented by the Base Hospital using a Field Advisory Report. Hospitals may not divert patients due to problems with staffing or hospital or Emergency Department (ED) bed availability.
- F. Destination for all Multi Casualty Incident (MCI) patients will be determined by the designated MCI base as outlined in Policy 6180 MCI Response.

### **IV. DESTINATION PROTOCOLS**

- A. Critical patients will be transported to the closest facility regardless of patient preference. Prehospital personnel will contact that hospital for any requested orders and for a patient advisory. If DGMC is the receiving facility, prehospital personnel will contact NBMC or VVH for physician orders.

- B. Non-critical patients may be transported to the receiving facility of the patient's choice. In the event that a non-critical patient being transported to DGMC requires physician orders, Paramedics will contact NBMC or VVH.
- C. In the event that a patient does not meet the definition of a critical patient but the evaluating Paramedic is concerned that the patient may be unstable for transport to a more distant facility, the paramedic is advised to discuss the case with the base physician at the facility of the patient's choice. The base hospital physician may then, if appropriate, override the patient's choice destination and direct the ambulance to the nearest receiving facility. In these cases, the base hospital will be responsible for communicating with the receiving facility.
- D. Patients requesting transport to receiving facilities outside of Solano County which are not the closest receiving facility may be accommodated only if all the following circumstances are met:
  - 1. The patient is non-critical;
  - 2. The patient's chosen facility is open to receive ambulances and is licensed as having Basic or Comprehensive Emergency Services and;
  - 3. The ambulance provider agency has the capability to provide the requested transport without adversely impacting any of their responsibilities regarding 911 system response.

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POLICY MEMORANDUM 6701

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**SUBJECT: TREATMENT AND REFERRAL FOR ILL PATIENTS DURING THE  
COVID-2019 OUTBREAK**

**AUTHORITY:**

California Health and Safety Code Sections 1797.94 and 1797.153  
California Code of Regulations Sections 101310

**PURPOSE:**

To guide the evaluation and transport decisions for patients experiencing mild respiratory illness possibly related to COVID-19.

**I. PRINCIPLES AND BACKGROUND**

The severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and its associated clinical syndrome known as COVID-19 is circulating worldwide and adversely impacting EMS everywhere, including in Solano County.

Based on the available data, most infected persons experience mild illness and fully recover. Those at high risk for severe illness and/or complications are patients who are elderly or have underlying medical conditions.

The Centers for Disease Control and Prevention (CDC) recommends that low risk patients potentially infected with COVID-19 experiencing mild disease symptoms self-isolate at home unless symptoms worsen.

## II. DEFINITIONS

A. Low risk potential COVID-19 patients: Patients greater than 12 months of age and less than 65 years of age with fever and/or respiratory complaints with NONE of the following complaints:

1. Shortness of breath or chest pain or syncope;
2. Abnormal vital signs for age except fever;
  - a. Normal vital signs are as follows:
    - i. Adult
      - SBP 100 – 199mm/Hg / DBP 60 – 90mm/Hg
      - Heart rate 60 – 100 beats per minute
      - Respiratory rate 12 – 20 breaths per minute
      - Oxygen saturation  $\geq 95\%$
      - Temperature  $96.8^{\circ} - 100.2^{\circ} \text{ F}$  or  $36^{\circ} - 37.9^{\circ} \text{ C}$
    - ii. Pediatric

| Age Range                 | HR<br>beats/min | RR<br>breaths/min | BP<br>Systolic | BP<br>Diastolic |
|---------------------------|-----------------|-------------------|----------------|-----------------|
| Newborn<br>0-1 month      | 100-180         | 30-60             | 73-92          | 52-65           |
| Infant<br>1-12 months     | 80-150          | 30-60             | 90-109         | 53-67           |
| Toddler<br>1-3 years      | 75-130          | 25-35             | 95-105         | 56-68           |
| Pre-School<br>3-5 years   | 75-120          | 22-32             | 99-110         | 55-70           |
| School Age<br>5-12 years  | 70-110          | 20-30             | 97-118         | 60-76           |
| Adolescent<br>13-18 years | 65-105          | 16-22             | 110-133        | 63-83           |

- Oxygen saturation  $\geq 95\%$
- Temperature  $96.8^{\circ} - 100.2^{\circ} \text{ F}$  or  $36^{\circ} - 37.9^{\circ} \text{ C}$

3. History of significant comorbid disease including chronic obstructive pulmonary disease (COPD), congestive heart failure (CHF), cardiac disease, renal failure immunodeficiency, critically ill appearing, or emergency medical condition per paramedic judgement.

B. Emergency Medical Condition: A condition or situation in which an individual has an immediate need for medical attention.

C. Self-isolation: The practice of a symptomatic person isolating themselves and completely avoiding contact with other people, in an effort to prevent transmitting an infection to other people.

### III. PROCEDURE WHEN ENCOUNTERING POTENTIAL LOW RISK COVID-19 PATIENTS

- A. Low-risk patients with mild symptoms or respiratory infection (e.g., fever, cough, upper respiratory illness) who appear non-critical should be advised that their current condition does not require transport to the Emergency Department (ED).
  - B. EMS personnel shall advise the patient or the patient's legal representative to:
    - 1. Self-isolate at home, avoid having guests to the home, apply appropriate social distancing, avoid contact with high-risk persons, and self-monitor their condition for worsening symptoms.
    - 2. Cover the nose and mouth when coughing and sneezing and practice frequent hand hygiene.
    - 3. Seek follow-up treatment as needed with their physician if their symptoms worsen.
    - 4. Continue isolation at home until patient is symptom free and cleared by the patient's physician based on current guidelines from the CDC and Solano County Public Health.
  - C. The advice given will be documented on the Patient Care Report (PCR). The following statement is recommended for use in documentation and patient advisement:

**"It appears that you do not require immediate care in the emergency department. You should seek care with your regular healthcare provider or a doctor's office or clinic if symptoms worsen. If you develop shortness of breath or other severe symptoms, recontact 9-1-1."**
  - D. If the patient or the patient's legal representative requests the patient be transported after assurance that transport is not needed, EMS personnel should honor the request and transport per Solano County EMS Policy 6700, Destination Protocols for Ambulances.
    - 1. EMS personnel transporting patients with suspected COVID-19 shall notify the receiving facility in advance prior to arrival and may be directed to a screening area other than the ED.
    - 2. EMS personnel who are assessing, treating or transporting a patient with suspected COVID-19 shall use appropriate PPE at all times to including gown, gloves, mask, and eye protection.

Patients with suspected COVID-19 should wear a surgical mask at all times during transport.
    - 3. Upon arrival at the ED and after transfer of care, EMS personnel should follow PPE doffing and hand hygiene procedures.
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**POLICY MEMORANDUM 6800**

**Implementation Date: May 21, 2023**  
**Review Date: March 1, 2025**

**REVIEWED/APPROVED BY:**

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**BENJAMIN GAMMON, EMT-P, EMS AGENCY ADMINISTRATOR**

**SUBJECT: PARAMEDIC UTILIZATION OF AUTOMATIC TRANSPORT  
VENTILATORS (ATV) DURING INTERFACILITY TRANSPORT (IFT)**

**AUTHORITY:** Health and Safety Code, Division 2.5, Sections 1797.220  
California Code of Regulations, Title 22, Chapter 4, Article 1, Section  
100145

**I. PURPOSE:**

- A. To provide parameters for paramedic utilization of ATVs during IFTs.

**II. PARAMEDIC IFT OPTIONAL SKILLS**

- A. Only the Solano County ALS Exclusive Operating Area (EOA) Provider may be authorized to utilize Paramedic IFT optional skills.
- B. Only appropriately trained Paramedics employed by the ALS EOA Provider may utilize Paramedic IFT optional skills.
- C. ALS EOA provider using ATVs shall follow the manufacture instructions or use, maintenance, cleaning, and regular testing of the device. At a minimum, ATVs shall undergo annual preventative maintenance testing and maintenance by qualified manufacturer's representative personal.
- D. Paramedics must be thoroughly trained and regularly retrained on the use of ATVs. Training shall occur once annually and shall be documented.

### III. ATV PROCEDURES

- A. Written transfer orders from the transferring physician shall be obtained prior to transport. These orders will be attached to the electronic Patient Care Report (ePCR). These orders shall include:
  - 1. Orders for maintaining and adjusting ventilations via ATV settings during transport.
  - 2. Telephone number where the transferring physician can be reached during transport.
  - 3. Orders for maintaining sedation with sedatives that are within the basic scope of practice for paramedics.
- B. Ventilator support must be regulated by an ATV familiar to the paramedic.
- C. If an ATV failure occurs and cannot be corrected, the paramedic shall discontinue use of the ATV, initiate ventilation by a bag-valve-mask device, and notify the transferring physician as soon as possible.
  - 1. All ATV failures shall be reported to the EMS Agency via a Field Advisory Report (FAR).
- D. Paramedics shall continually observe the patient and document patient response to any changes while the ATV is operational.
- E. Initial ATV settings and any subsequent changes shall be documented on the ePCR.
- F. The paramedic is responsible for all airway management and must frequently reassess tracheostomy/endotracheal tube placement, including after each patient movement.
- G. A non-invasive BP monitor device shall be utilized. Vital signs shall be monitored and documented every 15 minutes and immediately if there is any change in patient status or adjustment of the ATV setting. Vital signs shall include pulse oximetry and cardiac monitoring which shall be maintained throughout transport.
- H. Continuous waveform capnography shall be utilized during transport.
- I. The ventilator that the paramedic provider will be using must be able to match the existing ventilator settings, and shall include the following minimum device features (including circuit):
  - 1. Modes:
    - a. Assist Control (AC).
    - b. Synchronized Intermittent Mandatory Ventilation (SIMV).
    - c. Controlled Mechanical Ventilation (CMV).
  - 2. Set rate of ventilations.
  - 3. Adjustable delivered tidal volume.
  - 4. Adjustable FiO<sub>2</sub>.
  - 5. Positive End-Expiratory Pressure (PEEP).

6. Adjustable Inspiratory and Expiratory ratios (I:E ratio).
7. Peak airway pressure gauge.
8. Alarms:
  - a. Peak airway pressure.
  - b. Disconnect.

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**POLICY MEMORANDUM 6801**

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**SUBJECT: PARAMEDIC MONITORING OF BLOOD TRANSFUSIONS DURING INTERFACILITY TRANSFERS (IFT)**

**AUTHORITY:** Health and Safety Code, Division 2.5, Sections 1797.220  
California Code of Regulations, Title 22, Chapter 4, Article 1, Section 100145

**I. PURPOSE:**

- A. To provide parameters for paramedic monitoring of blood transfusions during IFTs.

**II. PARAMEDIC IFT OPTIONAL SKILLS**

- A. Only the Solano County ALS Exclusive Operating Area (EOA) Provider may be authorized to utilize Paramedic IFT optional skills.
- B. Only appropriately trained Paramedics employed by the ALS EOA Provider may utilize Paramedic IFT optional skills.
- C. Patients will have pre-existing blood transfusions started at least 15 minutes prior to arrival at the transferring facility in peripheral or central IV lines. Prehospital personnel will not initiate blood transfusions.
  - 1. If the transfusion is started upon the arrival of transport, the transport will be delayed for 15 minutes to observe for any transfusion reactions.

### III. BLOOD TRANSFUSION PROCEDURES

- A. All patients will be maintained on a cardiac monitor and a non-invasive blood pressure monitor.
- B. Written transfer orders from the transferring physician shall be obtained prior to transport. These orders will be attached to the electronic Patient Care Report (ePCR). These orders shall include:
  - 1. Orders for maintaining and adjusting blood transfusion rate during transport.
  - 2. Telephone number where the transferring physician can be reached during transport.
- C. Paramedic personnel must be knowledgeable in the operation of the specific blood delivery/warming device.
- D. Regulation of the transfusion rate will be within the parameters defined by the transferring physician.
- E. Verify the patient and blood with the sending RN by checking the patient ID band against the blood label(s) and blood order for name, blood type and unit identifying number.
- F. Vital signs will be monitored and documented every 15 minutes and immediately if there is any change in patient status or change in transfusion rate.
- G. Monitor the patient for any signs and symptoms of a transfusion reaction. Monitor temperature for adverse effects if transport time exceeds 15 minutes. The following are the most common types of transfusion reactions that may occur:
  - 1. Hemolytic reactions: Hemolytic reactions are the most life-threatening. Clinical manifestations may vary considerably: fever, headache, chest or back pain, pain at infusion site, hypotension, nausea, generalized bleeding or oozing from surgical site, shock. The most common cause is from ABO incompatibility due to a clerical error or transfusion to the wrong patient. Chances of survival are dose dependent therefore it is important to stop the transfusion immediately if a hemolytic reaction is suspected. Give a fluid challenge.
  - 2. Febrile non-hemolytic reaction: Chills and fever (rise from baseline temperature of 1°C or 1.8°F). Document and report to hospital on arrival.
  - 3. Allergic reaction: Characterized by appearance of hives and itching.
  - 4. Anaphylaxis: May occur after administration of only a few mL's of a plasma containing component. Symptoms include coughing, bronchospasm, respiratory distress, vascular instability, nausea, abdominal cramps, vomiting, diarrhea, shock, and loss of consciousness.

5. Volume overload: Characterized by dyspnea, headache, peripheral edema, coughing, frothy sputum or other signs of congestive heart failure occurring during or soon after transfusion. Restrict fluid.
- H. If a suspected transfusion reaction occurs:
1. Interrupt the transfusion immediately.
  2. Contact the transferring hospital physician.
  3. Consult appropriate treatment protocol.
  4. Document any suspected transfusion reactions.
  5. Report to hospital staff immediately upon arrival.
- I. The paramedic shall document on the ePCR the total volume infused throughout the duration of the transport.

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**POLICY MEMORANDUM 6802**

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**SUBJECT: PARAMEDIC MONITORING OF MAGNESIUM SULFATE,  
NITROGLYCERIN, HEPARIN, AND/OR AMIODARONE INFUSIONS  
DURING INTERFACILITY TRANSPORTS (IFT)**

**AUTHORITY:** Health and Safety Code, Division 2.5, Sections 1797.220  
California Code of Regulations, Title 22, Chapter 4, Article 1, Section  
100145

**I. PURPOSE**

- A. To provide parameters for paramedic monitoring of magnesium sulfate, nitroglycerin (NTG), heparin, and/or amiodarone infusions during IFTs.

**II. PARAMEDIC IFT OPTIONAL SKILLS**

- A. Only the Solano County ALS Exclusive Operating Area (EOA) Provider may be authorized to utilize Paramedic IFT optional skills.
- B. Only appropriately trained Paramedics employed by the ALS EOA Provider may utilize Paramedic IFT optional skills.
- C. Patients will have pre-existing infusions in peripheral or central IV lines. Paramedics will not initiate magnesium sulfate, nitroglycerin, heparin, &/or amiodarone infusions.

- D. Magnesium sulfate, nitroglycerin, heparin, and/or amiodarone infusions will have been running for at least 10 minutes prior to transport.
- E. Patients will have maintained stable vital signs for the previous 30 minutes, and will not have more than two (2) medication infusions running exclusive of potassium chloride concentrations authorized under the paramedic basic scope of practice.
- F. The timeframes listed in items D and E above will not apply to patients who require immediate transport for critical interventions, when the transferring physician determines that immediate transport is necessary.

### **III. INFUSION PROCEDURES**

- A. All patients shall be maintained on a cardiac monitor and a non-invasive blood pressure monitor.
- B. Written transfer orders from the transferring physician shall be obtained prior to transport. These orders will be attached to the electronic Patient Care Report (ePCR). These orders shall include:
  - 1. Orders for maintaining and adjusting infusion rate during transport.
  - 2. Telephone number where the transferring physician can be reached during transport.
  - 3. Type of solution, dosage, and rate of infusion.
- C. Patients will be hemodynamically stable at time of transport.
- D. If medication administration is interrupted, the paramedic may restart the infusion as delineated in the transfer orders.
- E. All infusions, except for potassium chloride concentrations authorized under the paramedic basic scope of practice, will be in the form of an IV piggyback monitored by a mechanical pump familiar to the paramedic. In cases of pump malfunction that cannot be corrected, the infusion shall be discontinued and the transferring physician notified as soon as possible.
  - 1. If this occurs, a Field Advisory Report (FAR) will be submitted to the EMS Agency.
- F. The paramedic shall document on the PCR the total volume infused throughout the duration of the transport.

### **IV. MAGNESIUM SULFATE INFUSION PARAMETERS**

- A. Regulation of the infusion rate will be within parameters defined by the transferring physician.
- B. If the patient develops signs/symptoms of magnesium toxicity, the medication drip shall be discontinued and the transferring physician will be notified as soon as possible. Signs/symptoms of magnesium toxicity include:
  - 1. Thirst
  - 2. Diaphoresis

3. Hypotension
  4. Flaccid paralysis
  5. Respiratory depression
  6. Circulatory depression or collapse
  7. CNS depression
  8. Urine output <30mL/hr
  9. Chest pain or pulmonary edema
  10. Deep tendon reflexes (DTR) – depressed or absent
- C. Vital signs, including DTR's, shall be monitored and documented every 15 minutes and immediately if there is any change in patient status or medication adjustment.

**V. NTG INFUSION PARAMETERS**

- A. Infusion fluid will be D5W.
- B. Regulation of the infusion rate will be within parameters defined by the Transferring physician, but in no case will changes be greater than 10mcg/minute increments every 5-10 minutes. In cases of severe hypotension, the medication drip will be discontinued and the transferring physician will be notified as soon as possible.
- C. Discuss with transferring physician concomitant use of analgesics during transport (i.e., morphine sulfate, fentanyl).
- D. Vital signs shall be monitored and documented every 15 minutes and Immediately if there is any change in patient status or medication adjustment.

**VI. HEPARIN INFUSION PARAMETERS**

- A. Infusion fluid will be D5W or NS.
- B. Infusion rates shall be verified with the sending RN following changeover to The mechanical EMS transport pump, and will remain constant during transport. No regulation of the rate will be performed by the paramedic except to turn off the infusion completely.
- C. Vital signs shall be monitored and documented every 15 minutes and Immediately if there is any change in patient status.

**VII. AMIODARONE INFUSION PARAMETERS**

- A. Infusion rates may vary between 0.25 – 1 mg/min.
- B. Infusion rates will remain constant during transport. No regulation of the rate will be performed by the paramedic except to turn off the infusion completely.

- C. Vital signs will be monitored and documented every 15 minutes and immediately if there is any change in patient status.
- D. Y-Injection incompatibility – The following will precipitate with amiodarone hydrochloride:
  - 1. Heparin.
  - 2. Sodium Bicarbonate.
- E. Amiodarone IV infusion monitoring is not approved for patients less than 14 years old without base physician contact.
- F. For infusions greater than one hour, amiodarone concentrations should not exceed 2mg/mL unless a central venous catheter is used.